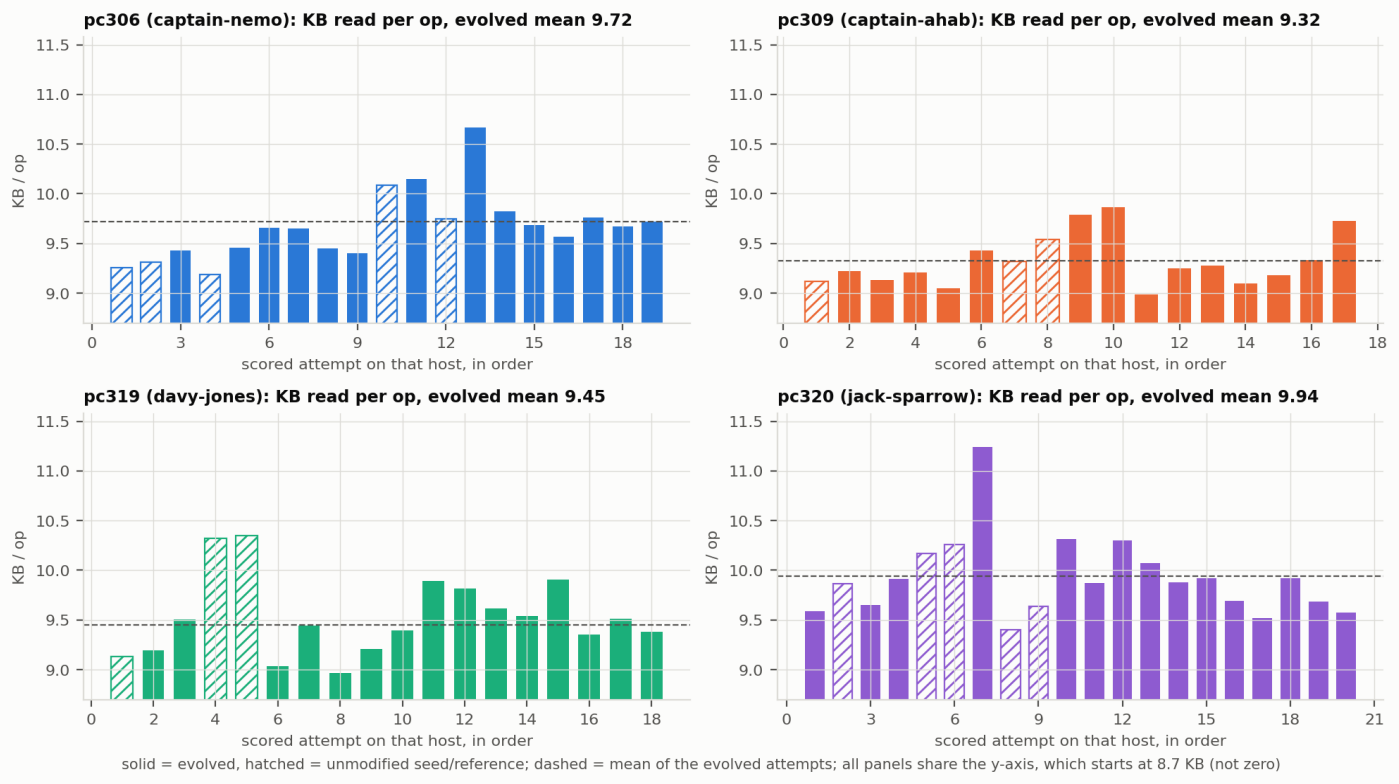
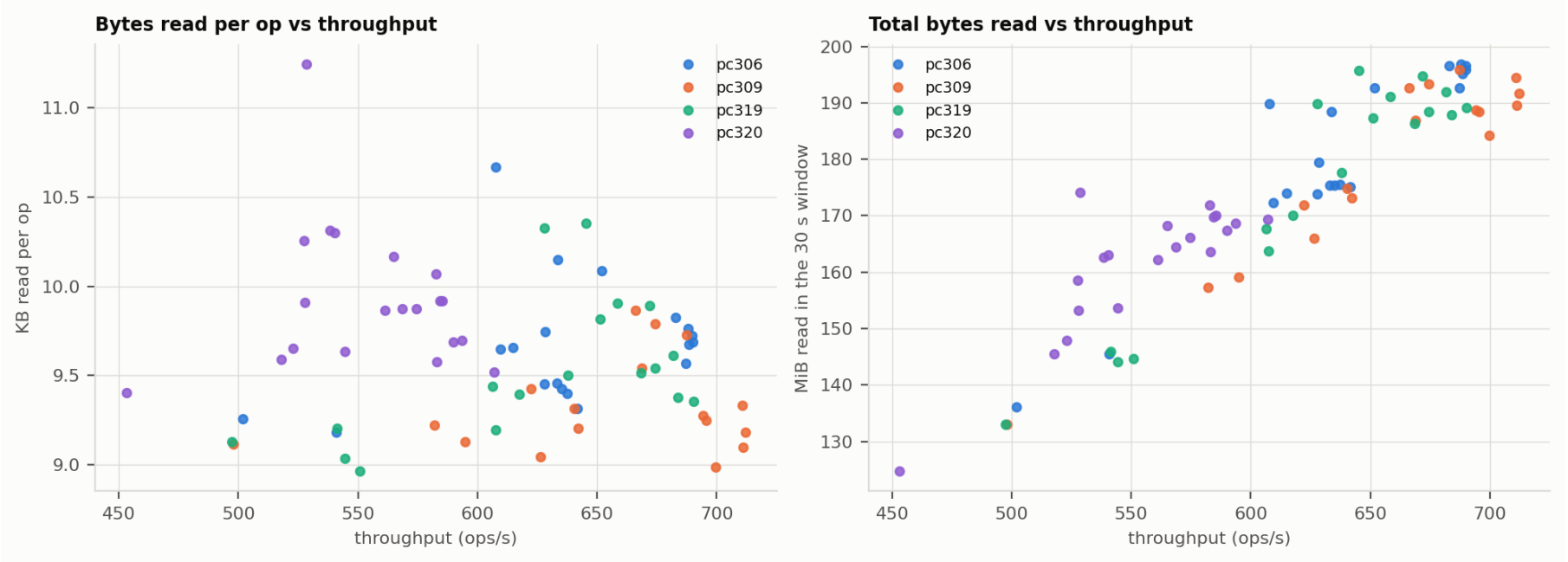
# Bytes read per operation — multi-host run 2026-10-21_003256

For every scored attempt (74 of 80; 6 crashed attempts have no measurement): bytes read per op = bytes read from disk in the measured phase ÷ operations completed in it, with operations = throughput × 30 s (the benchmark's measured window). Lower means each operation needed less disk I/O. One scored attempt = one evaluation on that agent's own host; hosts differ in speed, so compare within a host first.

| host (agent) | scored | KB/op: best by lowest | KB/op median | KB/op min – max | mean MiB read | unmodified seeds (KB/op) | best-throughput attempt: KB/op |
|---|---|---|---|---|---|---|---|
| pc306 (captain-nemo) | 19 | 9.18 (#4 9a36aaf7, 541 ops/s) | 9.66 | 9.18 – 10.67 | 180 | get_scan 9.26, s3fifo 9.31, sampling 9.18, mglru 10.09, noop 9.74 | 9.69 (3f1573c8, 690 ops/s) |
| pc309 (captain-ahab) | 17 | 8.99 (#11 c56fed27, 700 ops/s) | 9.25 | 8.99 – 9.87 | 179 | get_scan 9.12, noop 9.32, mglru 9.54 | 9.18 (7e6df4dc, 712 ops/s) |
| pc319 (davy-jones) | 18 | 8.97 (#8 281d22ad, 551 ops/s) | 9.47 | 8.97 – 10.35 | 175 | get_scan 9.13, fifo 10.32, mglru 10.35 | 9.35 (50b9f179, 690 ops/s) |
| pc320 (jack-sparrow) | 20 | 9.40 (#8 c63ebbfd, 453 ops/s) | 9.88 | 9.40 – 11.24 | 161 | noop 9.86, mglru 10.17, mru 10.26, lhd 9.40, s3fifo 9.64 | 9.52 (6b62bff1, 607 ops/s) |

Correlation of KB/op with throughput: -0.12 over all attempts; per host pc306 +0.28, pc309 +0.20, pc319 +0.54, pc320 -0.08. Note KB/op moves with the policy but is not what the run optimised: the score was throughput, and the best policy does not have the lowest KB/op.






## Every scored attempt

| host | # | commit | tput ops/s | MiB read | ops in 30 s | bytes/op | KB/op | design / title |
|---|---|---|---|---|---|---|---|---|
| pc306 | 1 | eb8d23ba =get_scan | 501.9 | 136.1 | 15057 | 9478 | 9.26 | baseline: reference/get_scan.c unchanged (scan-pid list + sampled LFU w/ index-block bonus) to calibrate my host |
| pc306 | 2 | bceae5c4 =s3fifo | 641.8 | 175.1 | 19252 | 9537 | 9.31 | baseline 2: reference/s3fifo.c unchanged. Context: get_scan.c scored 502 (2.4x below noop ~1215) on pc306; hypothes |
| pc306 | 3 | b17a7960 | 635.1 | 175.4 | 19053 | 9653 | 9.43 | s3fifo + LevelDB index-block pinning: last page of each file admitted straight to main w/ freq 3 and skipped by mai |
| pc306 | 4 | 9a36aaf7 =sampling | 540.8 | 145.5 | 16223 | 9405 | 9.18 | baseline 3: reference/sampling.c unchanged (sampled LFU k=20 + index bonus). Question: does pure frequency beat s3f |
| pc306 | 5 | daeac2fb | 633.2 | 175.4 | 18996 | 9682 | 9.46 | s3fifo knob 1/3: small queue 25% of cache (ref: cache/15 or main<=2*small). Hypothesis: 800-page small window too s |
| pc306 | 6 | 4ddceb13 | 614.9 | 174.0 | 18448 | 9890 | 9.66 | structural 1/3 readahead-aware s3fifo: eager promotion to main in folio_accessed (list_move) + never-accessed small |
| pc306 | 7 | 44700168 | 609.6 | 172.3 | 18287 | 9879 | 9.65 | structural 2/3 access-aware s3fifo: INVERT eval#5's lap rule. Eval#5 (lap for freq-0) = 615, bytes/IO unchanged (15 |
| pc306 | 8 | 5a7382d2 | 628.0 | 173.9 | 18839 | 9679 | 9.45 | structural 3/3 ablation: reference s3fifo + ONLY eager promotion (list_move to main in folio_accessed on 2nd hit). |
| pc306 | 9 | 1ff547d4 | 637.4 | 175.5 | 19122 | 9624 | 9.40 | s3fifo ghost 4x cache (ref 1x). Rationale: LevelDB block cache above page cache makes page-cache reuse distances lo |
| pc306 | 10 | b75f5c54 =mglru | 651.8 | 192.6 | 19554 | 10328 | 10.09 | reviewer: reference/mglru.c unchanged on pc306 to resolve 645 (davy) vs 565 (jack) cross-host split for identical c |
| pc306 | 11 | 7750f4f9 | 633.6 | 188.4 | 19009 | 10393 | 10.15 | structural 1/3 (retry, verifier fix) mglru+ghost-admission: refaulting folios added to youngest gen. Prior eval fai |
| pc306 | 12 | 0c4616da =noop | 628.4 | 179.4 | 18853 | 9978 | 9.74 | reviewer: unmodified noop.c on pc306 to anchor my host. ahab: noop=640 on pc309, jack: 561 on pc320 -> hosts differ |

| host | # | commit | tput ops/s | MiB read | ops in 30 s | bytes/op | KB/op | design / title |
|---|---|---|---|---|---|---|---|---|
| pc306 | 13 | 53c00e6d | 607.7 | 189.9 | 18232 | 10922 | 10.67 | mglru lane 2/3: reference mglru (652; noop on pc306 = 628) with new folios added at TAIL of gen list instead of hea |
| pc306 | 14 | 05ad47b0 | 683.0 | 196.6 | 20490 | 10061 | 9.83 | reviewer replication: davy-jones f578a1ba region-frequency CLOCK (672 on pc319) UNCHANGED on pc306. My pc306 anchor |
| pc306 | 15 | 3f1573c8 | 690.0 | 195.8 | 20700 | 9918 | 9.69 | region-CLOCK grace sweep 1/2: davy-jones f578a1ba (683 on pc306) with GRACE_NS 1s -> 200ms. Motivation: mglru head- |
| pc306 | 16 | 51067b5b | 687.2 | 192.6 | 20617 | 9796 | 9.57 | region-CLOCK combo: GRACE_NS 200ms (mine, 690 on pc306 vs 683 at 1s) + REGION_SHIFT 3 (davy-jones best on pc319: 68 |
| pc306 | 17 | 4173866b | 688.0 | 196.8 | 20642 | 9997 | 9.76 | region-CLOCK grace sweep 2/2: GRACE_NS 50ms, REGION_SHIFT 4. pc306 so far: grace 1s=683, 200ms=690, 200ms+shift3=68 |
| pc306 | 18 | b9876718 | 688.5 | 195.1 | 20654 | 9905 | 9.67 | region-CLOCK DEDUPE_NS 2ms -> 20ms (untaken knob) on my best 3f1573c8 (shift4, grace 200ms, 690). Region counter ag |
| pc306 | 19 | 3d305d1f | 689.9 | 196.5 | 20697 | 9955 | 9.72 | variance repeat (final eval 20/20): 3f1573c8 unchanged except comment (region-CLOCK shift4, grace 200ms; first run |
| pc309 | 1 | ec1f1584 =get_scan | 497.9 | 133.0 | 14938 | 9336 | 9.12 | baseline: reference get_scan.c unchanged (scan-pid list + sampled LFU) to calibrate my host |
| pc309 | 2 | 0b0faad4 | 582.1 | 157.3 | 17464 | 9445 | 9.22 | structural 1/3 scan-aware S3-FIFO (compile fix 2: static size counters): scan-pid list evicted first (256-page rese |
| pc309 | 3 | a7c5ead2 | 594.9 | 159.1 | 17846 | 9348 | 9.13 | scan-aware S3-FIFO v2 (2/3): ghost ALL non-scan evictions (incl MAIN), ghost 64k entries storing freq; ghost hit -> |
| pc309 | 4 | f4670f97 | 642.2 | 173.2 | 19265 | 9427 | 9.21 | ablation 3/3: v2 with scan-pid special casing OFF (no scan list, scan accesses count) -> plain S3-FIFO + ghost-all |
| pc309 | 5 | c142f84a | 626.3 | 166.0 | 18790 | 9264 | 9.05 | structural 1/3 NEW-filtered S3-FIFO: never-accessed folios (GET readahead junk) in NEW list evicted first beyond 51 |
| pc309 | 6 | fe40227e | 622.4 | 171.9 | 18671 | 9654 | 9.43 | long-memory ghost: f4670f97 (642) with ghost 64k per-CPU-LRU -> 256k common LRU (~20x cache). Two-level-cache view: |
| pc309 | 7 | ad992e58 =noop | 640.4 | 174.8 | 19213 | 9540 | 9.32 | calibration on fast host pc309: unmodified noop.c (kernel default, comment-only change). Open question: is kernel d |
| pc309 | 8 | 21e921d8 =mglru | 668.8 | 186.9 | 20063 | 9768 | 9.54 | reviewer: same-host pair. reference/mglru.c unchanged on pc309, where noop = 640.4 (ad992e58). Other hosts: mglru 6 |
| pc309 | 9 | 0a062399 | 674.4 | 193.4 | 20233 | 10023 | 9.79 | mglru + access aging: halve folio access count (min 1) on each tier-protection promotion, like kernel folio_inc_gen |
| pc309 | 10 | 55e53c70 | 666.2 | 192.6 | 19988 | 10104 | 9.87 | structural 1/3 (verifier fix: hash i_ino not host ptr) MGLRU + persistent region tiers: tier from folio accesses + |
| pc309 | 11 | c56fed27 | 699.6 | 184.2 | 20988 | 9203 | 8.99 | region-CLOCK team-best 50b9f179 (690 pc319) + miss-cost weighting: GET-thread accesses +2, scan-thread +1 (scan_pid |
| pc309 | 12 | acb7b208 | 695.6 | 188.5 | 20867 | 9472 | 9.25 | same-host control: team-best 50b9f179 (region-CLOCK) unchanged (comment only) on pc309, to attribute my c56fed27 GE |
| pc309 | 13 | 32f8132e | 694.3 | 188.7 | 20830 | 9499 | 9.28 | region-CLOCK 50b9f179 with counter array 2^19 -> 2^21 slots (fewer region hash collisions inflating cold-region cou |
| pc309 | 14 | 0a187508 | 711.1 | 189.6 | 21334 | 9319 | 9.10 | region-CLOCK 50b9f179 + region-coherent eviction (jack-sparrow draft, last untested skill extension): evicting a fo |
| pc309 | 15 | 7e6df4dc | 712.2 | 191.6 | 21367 | 9403 | 9.18 | repeat of 0a187508 (region-CLOCK + region-coherent eviction, 711.2) unchanged except comment, to confirm +2.2% over |
| pc309 | 16 | db1edf33 | 710.9 | 194.4 | 21328 | 9558 | 9.33 | region-coherent eviction DOOM_NS 100ms -> 300ms (base 7e6df4dc confirmed 711.2/712.2 on pc309). Longer doom window |
| pc309 | 17 | 10ecff9e | 687.3 | 195.9 | 20620 | 9962 | 9.73 | final eval 20/20: region-coherent eviction (best 7e6df4dc, 711/712) with REGION_SHIFT 3 -> 4. Without doom shift 4 |
| pc319 | 1 | f3034a35 =get_scan | 497.3 | 133.0 | 14919 | 9348 | 9.13 | baseline: reference/get_scan.c unchanged (scan-pid list + sampled LFU) to calibrate my host |
| pc319 | 2 | 3c5a7dfe | 607.5 | 163.7 | 18226 | 9418 | 9.20 | scan-aware S3-FIFO (fixed ctx compile err): scan-pid folios -> scan list evicted first (promote to main if GET-touc |
| pc319 | 3 | 0a4d0eed | 637.9 | 177.6 | 19136 | 9732 | 9.50 | falsification A/B on s3fifo ref (642): scan-pid-inserted folios admitted straight to main (opposite of scan-first e |

| host | # | commit | tput ops/s | MiB read | ops in 30 s | bytes/op | KB/op | design / title |
|---|---|---|---|---|---|---|---|---|
| pc319 | 4 | 77a39eb0 =fifo | 627.8 | 189.9 | 18835 | 10572 | 10.32 | diagnostic: reference/fifo.c unchanged (no accessed hook, no maps). Bytes/op is ~9.1-9.6KB across ALL attempts (mis |
| pc319 | 5 | 57af07f9 =mglru | 645.3 | 195.7 | 19358 | 10600 | 10.35 | reference/mglru.c unchanged (untested reference). Context: fifo 628 ~= s3fifo 642 > noop 561 -> hit-ratio quality w |
| pc319 | 6 | 925ce075 | 544.4 | 144.1 | 16331 | 9252 | 9.04 | structural 1/3 history-LFU (retry; prev fail = 'struct slot' clashes with vmlinux.h, renamed): single list, sampled |
| pc319 | 7 | 7ff71cd0 | 606.4 | 167.7 | 18192 | 9666 | 9.44 | s3fifo + consumed-scan-first: scan-inserted folios move to a consumed list on the scan's first READ and that list i |
| pc319 | 8 | 281d22ad | 550.9 | 144.7 | 16526 | 9181 | 8.97 | structural 2/3 history-LFU: v1 (544, 925ce075) + count at most one access per page per 2ms (scan thread reads many |
| pc319 | 9 | d04239a3 | 541.4 | 146.0 | 16241 | 9426 | 9.21 | structural 3/3 history-LFU: v2 (551) + 1s grace for newly inserted folios (score 2^40+count). Hypothesis: throughpu |
| pc319 | 10 | f2ca3e81 | 617.5 | 170.0 | 18526 | 9622 | 9.40 | history-LFU via ITERATE not SAMPLE: same persistent freq sketch + 1s grace as 541-551 versions, but eviction = CLOC |
| pc319 | 11 | f578a1ba | 672.0 | 194.7 | 20159 | 10127 | 9.89 | structural 1/3 region-frequency CLOCK: f2ca3e81 (617, history-LFU via iterate) with the persistent counter keyed by |
| pc319 | 12 | 38ee907f | 651.3 | 187.3 | 19538 | 10052 | 9.82 | structural 2/3 region-frequency CLOCK: REGION_SHIFT 4->5 (128KB regions = default readahead window). Parent f578a1b |
| pc319 | 13 | 75eae97c | 681.8 | 192.0 | 20453 | 9843 | 9.61 | structural 3/3 region-frequency CLOCK: REGION_SHIFT 3 (32KB). Sweep so far: shift4(64KB)=672 f578a1ba, shift5(128KB |
| pc319 | 14 | 2edd0685 | 674.3 | 188.5 | 20230 | 9771 | 9.54 | region-frequency CLOCK sweep: REGION_SHIFT 2 (16KB). So far shift0 617, shift3 682, shift4 672, shift5 651 (pc319) |
| pc319 | 15 | 7947fa05 | 658.5 | 191.1 | 19755 | 10143 | 9.91 | region-freq CLOCK shift3 (682) + CLOCK pass thresholds 1/3/8 -> 2/6/16 (region counters aggregate 8 pages so counts |
| pc319 | 16 | 50b9f179 | 690.3 | 189.2 | 20710 | 9580 | 9.35 | region-freq CLOCK: nemo combo 51067b5b (shift3 + grace 200ms, 687 pc306) + DEDUPE 2ms->20ms (region count = number |
| pc319 | 17 | 2e5d5053 | 668.4 | 186.3 | 20053 | 9742 | 9.51 | region-freq CLOCK 50b9f179 (690) with LOWER CLOCK thresholds 1/3/8 -> 0/1/3 (raising to 2/6/16 lost 682->658, so te |
| pc319 | 18 | 6a2eaff5 | 683.9 | 187.9 | 20516 | 9604 | 9.38 | final: resubmit of 50b9f179 (region-freq CLOCK, 690.3) with a comment-only change, to measure same-host run-to-run |
| pc320 | 1 | 48ab7261 | 517.8 | 145.5 | 15534 | 9821 | 9.59 | v1 structural 1/3: scan-first FIFO drain + CLOCK(2-bit freq) general list, promote scan pages touched by GET thread |
| pc320 | 2 | 897780ff =noop | 561.3 | 162.2 | 16838 | 10101 | 9.86 | calibration: unmodified noop.c (kernel default reclaim) on my host, to check whether ~500 ops/s BPF policies are be |
| pc320 | 3 | 392a37e1 | 523.0 | 147.9 | 15690 | 9884 | 9.65 | v2: scan-only minimal-overhead policy (no per-folio map, no folio_accessed); evict scan-pid pages FIFO first, propo |
| pc320 | 4 | 8cd537d2 | 527.9 | 153.3 | 15838 | 10149 | 9.91 | v3: v2 scan-only with SCAN_MIN_PAGES 32->2048 (8MiB reserve). Tests readahead-thrash hypothesis: rios ~constant acr |
| pc320 | 5 | 212162dd =mglru | 564.9 | 168.3 | 16948 | 10413 | 10.17 | lane change (scan-first lane closed: reserve 32->2048 gave 523->528, all scan-first policies < noop 561 < s3fifo 64 |
| pc320 | 6 | 30244447 =mru | 527.5 | 158.5 | 15825 | 10502 | 10.26 | hypothesis test: LevelDB block cache sits above page cache, so page-cache hit signal is inverted (2nd-level cache). |
| pc320 | 7 | 123b7e87 | 528.6 | 174.1 | 15858 | 11512 | 11.24 | falsification (mirror of my scan-first v3=528): GET-FIRST two-FIFO - GET pages evicted first beyond 2048-page reser |
| pc320 | 8 | c63ebbfd =lhd | 453.1 | 124.8 | 13592 | 9628 | 9.40 | HOST CALIBRATION WARNING: pc320 (mine) is ~12% slower than pc319: mglru.c = 565 here vs 645 on davy's pc319; my noo |
| pc320 | 9 | c2899433 =s3fifo | 544.5 | 153.7 | 16334 | 9867 | 9.64 | REVIEW/falsification: does s3fifo actually beat kernel default? Only I have measured noop (561 on pc320); team comp |
| pc320 | 10 | 20218501 | 538.4 | 162.7 | 16153 | 10562 | 10.31 | mglru-tune 1/3 (engineer lane: MGLRU is best within-host on pc306 651.8 vs s3fifo 641.8 and on pc320 565 vs s3fifo |
| pc320 | 11 | e61194f0 | 574.5 | 166.2 | 17235 | 10111 | 9.87 | mglru-tune 2/3: tune1 (tail insert) LOST 565->538, so evicting newest-inserted never-accessed folios (accesses==1, |

| host | # | commit | tput ops/s | MiB read | ops in 30 s | bytes/op | KB/op | design / title |
|---|---|---|---|---|---|---|---|---|
| pc320 | 12 | 7a6652e4 | 540.3 | 163.0 | 16209 | 10545 | 10.30 | mglru-tune 3/3: tune2 (always-oldest-gen insert, 574.5, my best vs ref 565) + refaulted folios whose ghost tier>=1 |
| pc320 | 13 | 10e9a948 | 582.7 | 171.9 | 17480 | 10312 | 10.07 | cross-host replication: davy-jones f578a1ba region-frequency CLOCK (672 on pc319) unchanged (comment only) on pc320 |
| pc320 | 14 | e5fa0ac1 | 568.5 | 164.5 | 17056 | 10113 | 9.88 | region-freq CLOCK (davy f578a1ba; replicated 582.7 on pc320 = +3.1% over mglru 565) + grace only when region count> |
| pc320 | 15 | 2a05859b | 584.3 | 169.8 | 17529 | 10157 | 9.92 | region-freq CLOCK knob nobody is sweeping: AGE_SHIFT 18->15 (8x faster count halving; at 2^18 counts barely decay w |
| pc320 | 16 | 922e509b | 593.6 | 168.6 | 17807 | 9928 | 9.70 | cross-host replication of team-best config: captain-nemo 51067b5b (region-freq CLOCK, REGION_SHIFT 3, GRACE 200ms; |
| pc320 | 17 | 6b62bff1 | 607.1 | 169.3 | 18213 | 9747 | 9.52 | region-freq CLOCK pass thresholds (1,3,8) -> (2,4,16) on team-best nemo 51067b5b (593.6 on pc320). Untouched knob: |
| pc320 | 18 | 9392a608 | 585.3 | 170.1 | 17558 | 10158 | 9.92 | thresholds trend test: (2,4,16)=607.1 > (1,3,8)=593.6 on pc320; now (3,6,24) on nemo 51067b5b base |
| pc320 | 19 | e7b0c611 | 589.9 | 167.4 | 17697 | 9919 | 9.69 | confirmation re-run of 6b62bff1 thresholds (2,4,16) (607.1). Neighbors on pc320: (1,3,8)=593.6, (3,6,24)=585.3 |
| pc320 | 20 | 64eef3e1 | 583.1 | 163.6 | 17493 | 9807 | 9.58 | final eval 20/20: cross-host replication of team best captain-ahab c56fed27 (699.6 pc309; region-CLOCK + GET-access |